

Marketing for Engineers

Washington Accord Graduate Requirements:

- Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

Professional practice module – topics

1. Course introduction & professionals
2. Law
3. Intellectual property
4. Marketing for engineers (today)
 - What is marketing?
 - Marketing & engineers
 - The marketing mix: the 4 p's
5. Professional engineering practice

What is Marketing?

Advertising and selling

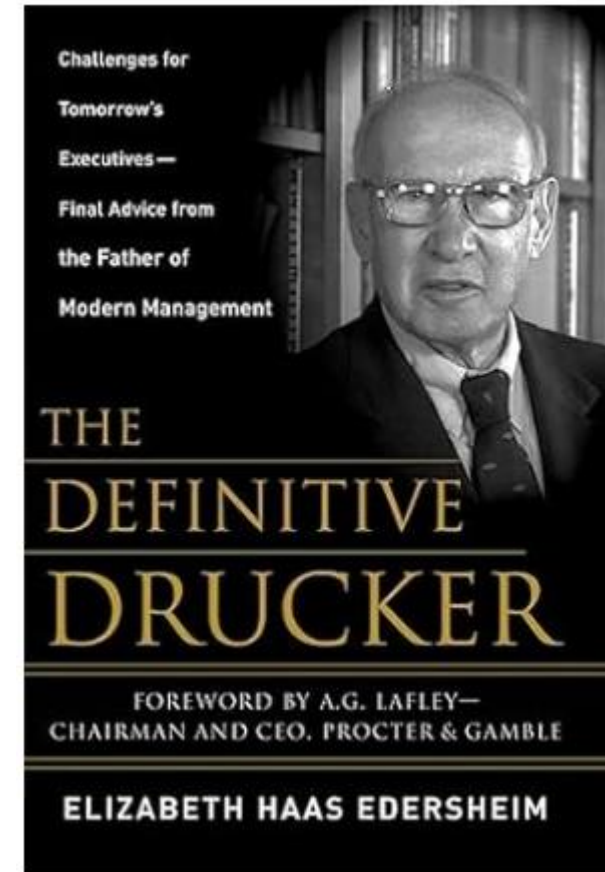
OR

A social and managerial process by which individuals and groups obtain what the **need** and **want** through creating and exchanging products and value with others

Marketing

“The aim of marketing is to make selling superfluous. The aim is to know and understand the customer so well that the product or service fits and sells itself.”

- Peter Drucker



Mistake 1: Product Orientation

Design Engineering

It's a good thing, so it will sell



Build a better mousetrap and the world will beat a path to your door...

Better mousetrap





\$6.00



Easy Trap Catcha Wooden
Mouse Trap 2 Pack



Intruder Rodent Control
Mouse Trap

1pack \$6.50 / 1ea

\$6⁵⁰

Mistake 2: Production Orientation

If we produce it more efficiently than the others can, it will sell



© Henry Ford Museum

“You can have any colour as long as it is black”

Marketing Orientation

If we solve **the customer's problem** better than anyone else,
and **let them know** we can, it will sell

Marketing is about solving customers' problems
Engineering is about solving customers' problems

What does this mean to the (entrepreneurial) engineer?

The biggest mistake you can make in a company:

Build something that no-one wants

Focus on Customer Needs and Wants

“Personally, I am very fond of strawberries and cream, but have found that for some strange reason, fish prefer worms.”

- Dale Carnegie

“If I had asked people what they wanted, they would have said faster horses.”

- Henry Ford

The Value Proposition

Who wants it and what will they pay for it?



The Marketing World

- You **must** understand your customer!
- Who is going to use your product?
- What are their needs?
- What is the competitive landscape?

The Consumer (Customer)

- Who is the customer?
 - Where are they?
 - How much money do they have?
 - What drives them?
- Split customers into different groups – *market segmentation*:
 - Demographic
 - Geographic
 - Behavioural
 - Psychographic

What drives consumers?

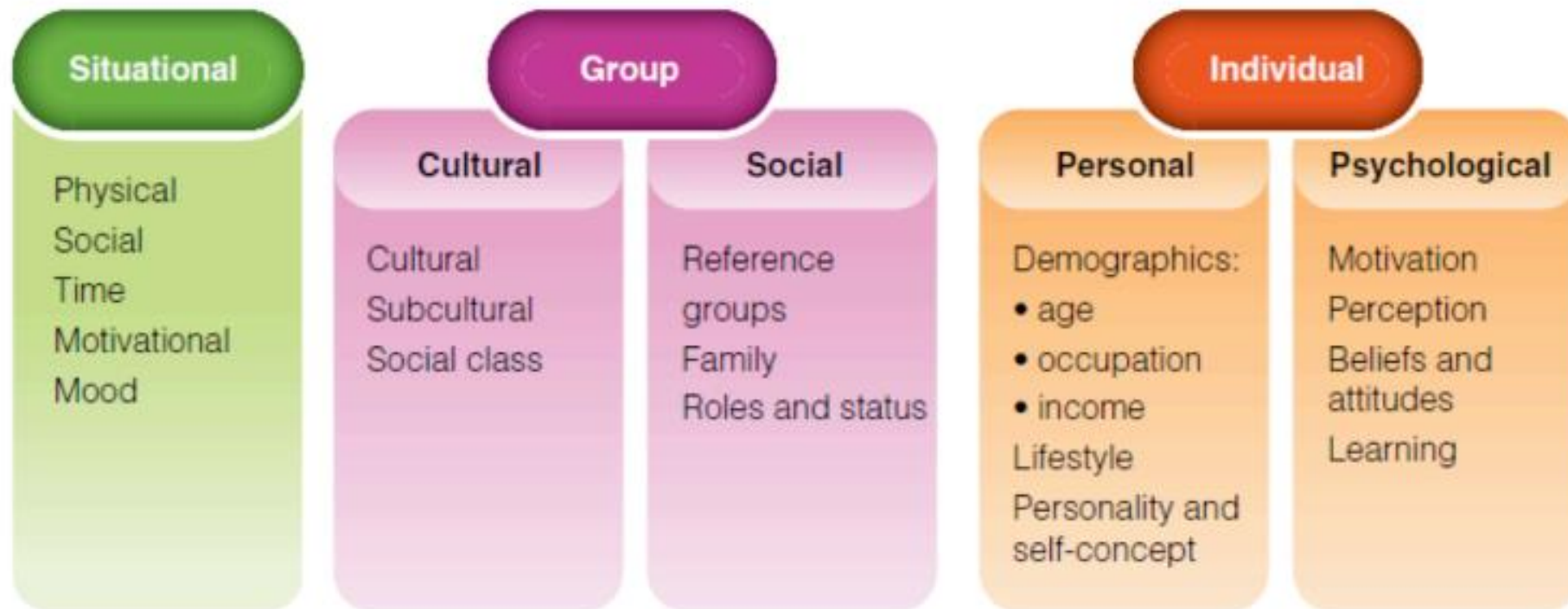
- Problem solving
 - Consumers buy to solve problems
- Psychoanalytical
 - Consumers are emotive buyers
- Cognitive miser
 - Consumers take the easiest option
- Collective decision maker
 - Consumers buy what everyone else is buying
- Economic
 - Consumers are rational buyers, who buy the best value option

But consumers are not rational...



© BBC

Consumer influences



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Marketing Strategy

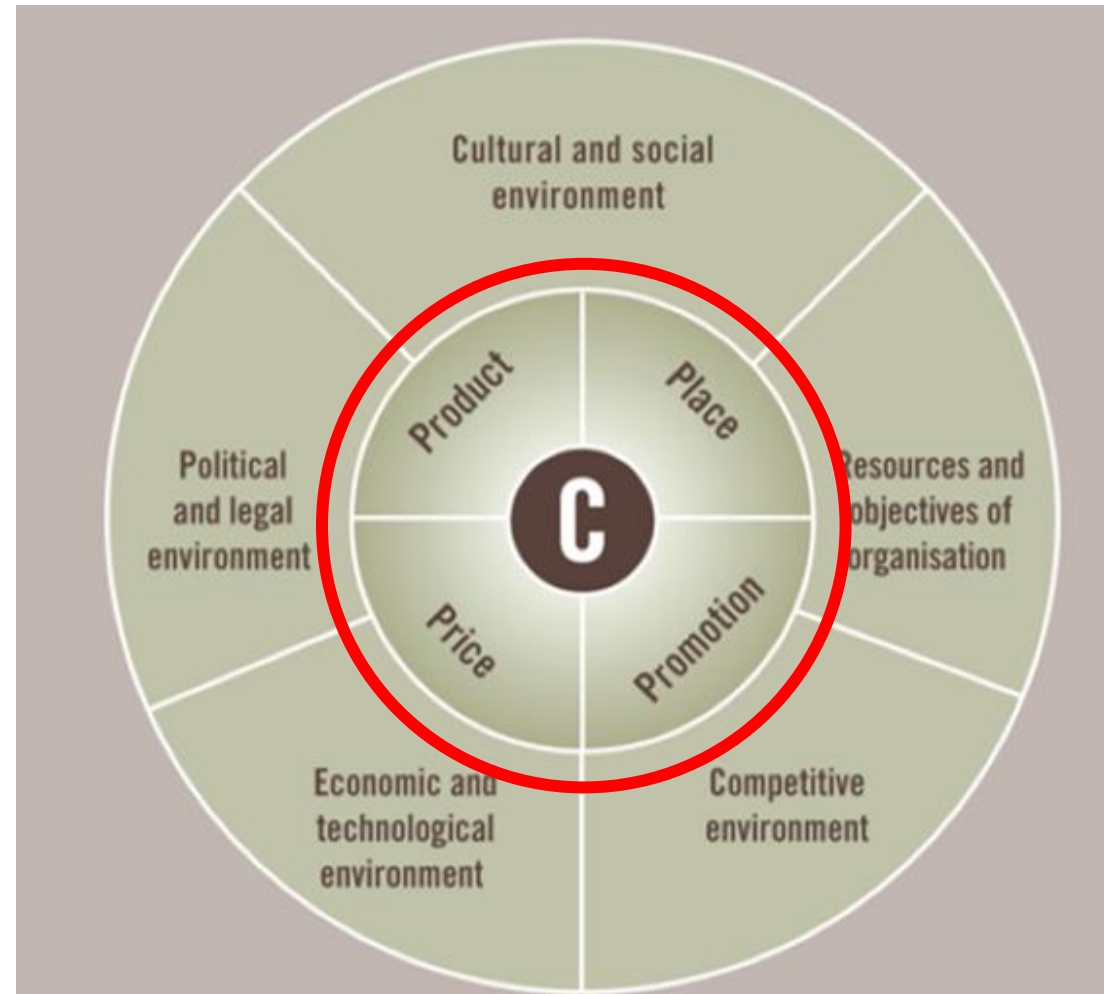
- A business needs information about:

1. The macro environment

2. The micro environment

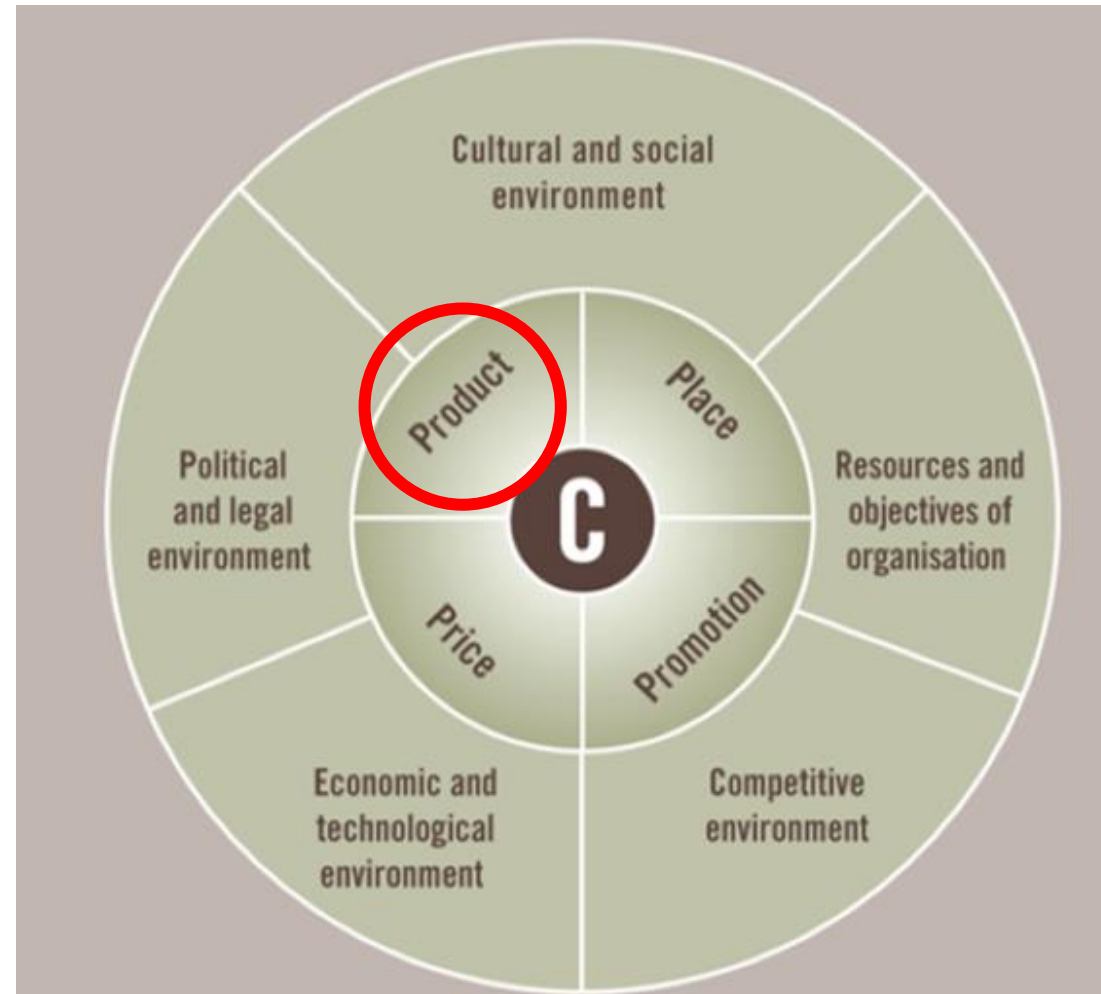
3. The customer

Marketing Strategy: The Marketing Mix



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The Marketing Mix: Product



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Positioning and Differentiation

- Positioning: how consumers *think* of your product compared to similar ones
 - May not be based on objective differences!
 - Influenced by marketing – brand perception
- Differentiation – how your product is *actually* different compared to similar ones
 - Can be between competitors or your own products

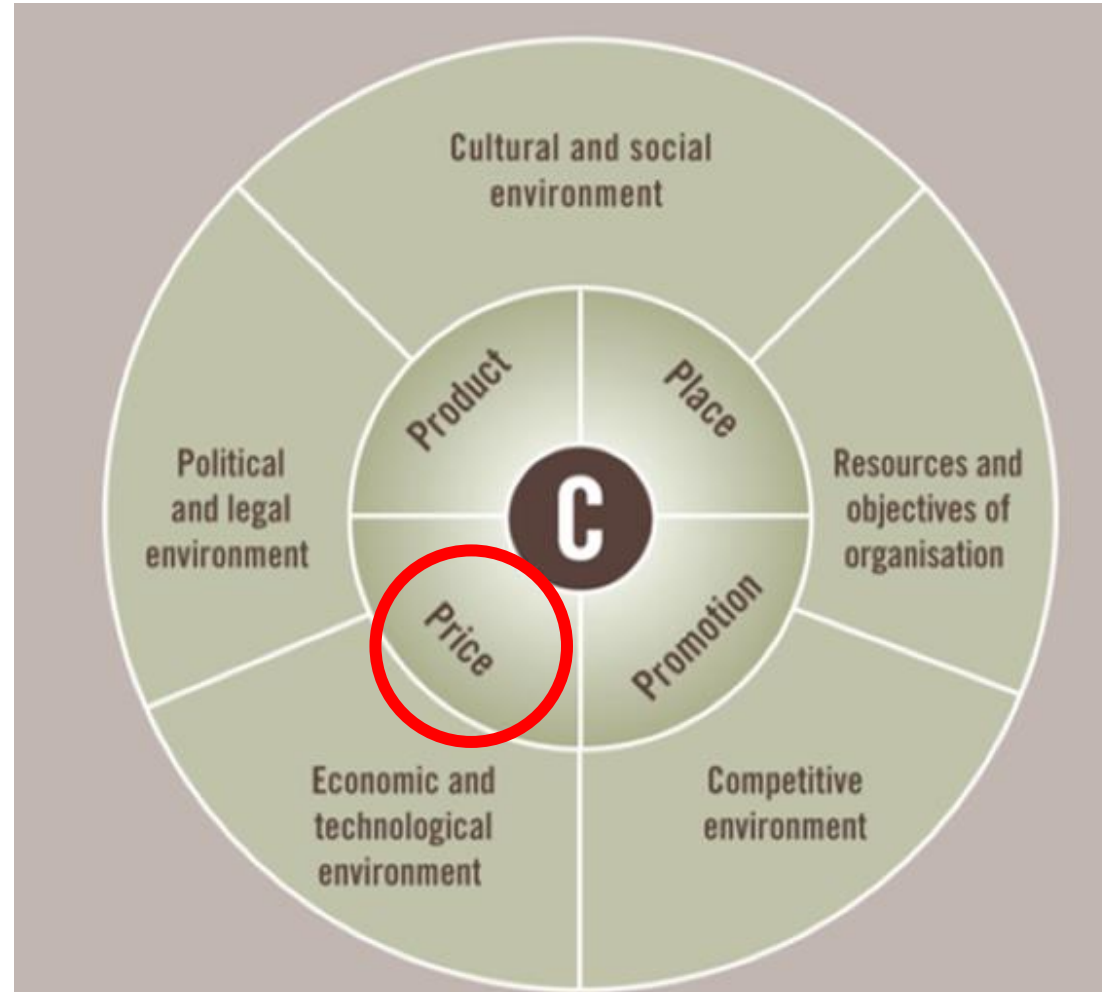
Differentiation

- What makes your product different?
 - What is your Unique Selling Point (USP)?
- Vertical differentiation: measurable differences between products
 - Faster, higher quality, longer life,...
 - Ranking vs. price – low quality may not be a competitive disadvantage!
- Increasing one feature may harm others – what do your customers want most?

Differentiation

- Horizontal differentiation: subjective differences between products
 - Colour, flavour, packaging...
- What does your market segment like?
 - Can you increase your market share by adding more choices?
- Can also differentiate by location, target demographic, technological advantage etc.

The Marketing Mix: Price

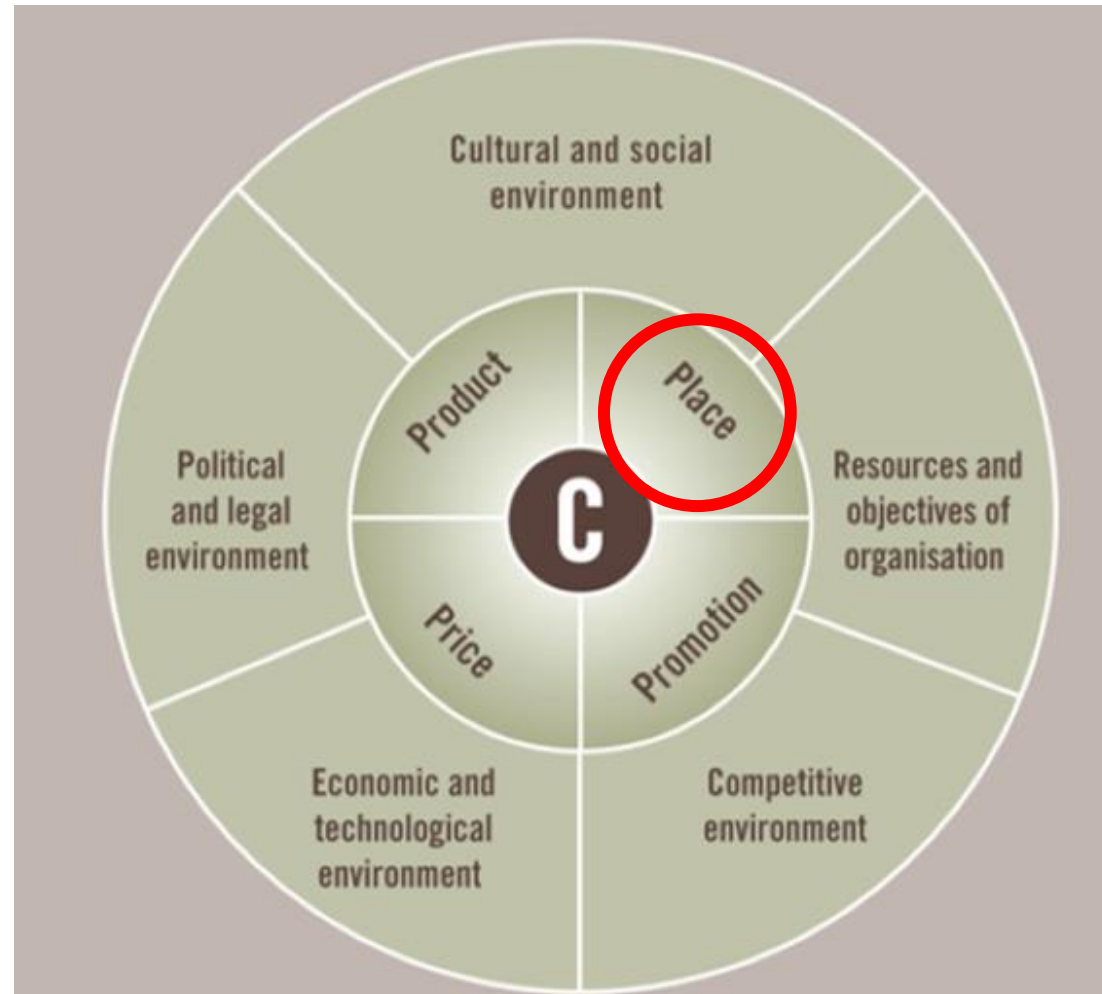


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Pricing Strategies

- Competitive pricing
- Cost plus pricing
- Penetration pricing
- Price skimming
- Value pricing
- Psychological pricing
- Freemium

The Marketing Mix: Place



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Not just where, but how and when (distribution channels, warehousing)



Coca Cola Amatil
distribution centre

Place – Vending Machines



Place – Online

 Coca Cola Soft Drink Coke 1.5L \$0.32 / 100mL \$4⁷⁹ Add to trolley	Special 20% Off  Coca Cola Soft Drink 330ml cans 30pack \$0.26 / 100mL \$26⁰⁰ Was \$34.00 Save \$8.00 Add to trolley	Member Price  Coca Cola Soft Drink Coke 2.25L \$0.17 / 100mL \$3⁸⁰ Non-member \$5.29 Add to trolley
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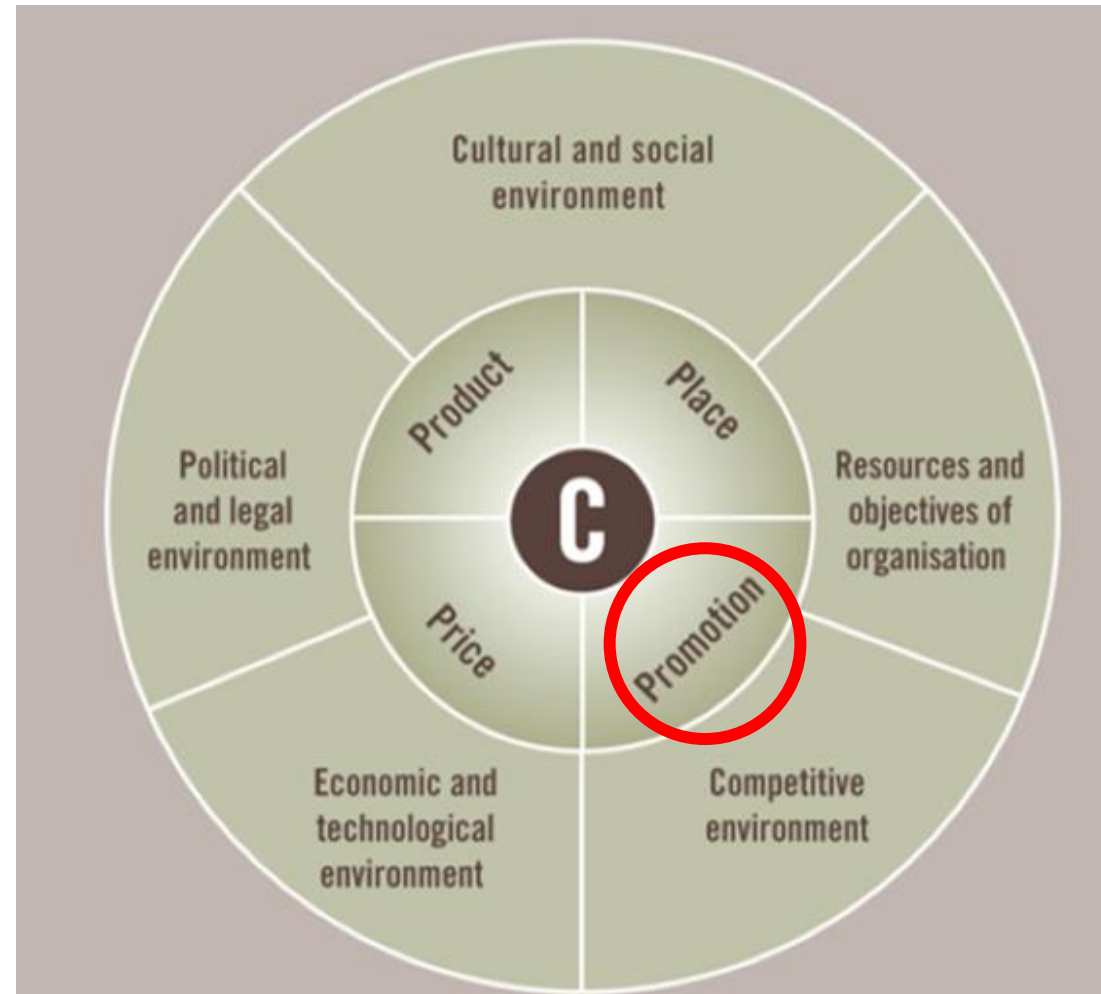
Compare to ~\$4
for 300ml at a
foodcourt

Place – Exclusive



© Capitol Signs

The Marketing Mix: Promotion



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Product: Milk



*Creator: Ajit Gangawane / EyeEm
Credit: Getty Images/EyeEm*

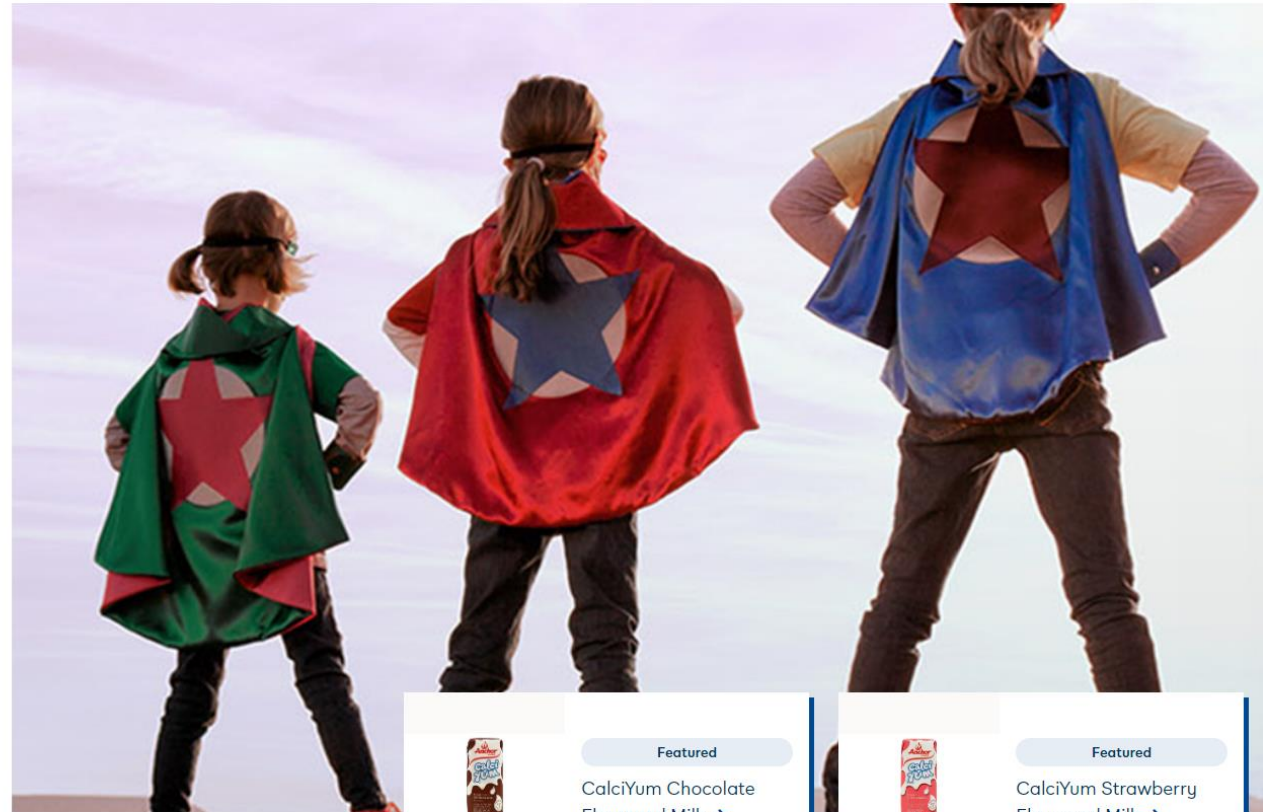
Flavoured Milk

[Products](#)[Recipes](#)[Dairy Nutrition](#)[Caring for Communities](#)[Caring for the Land](#)[About Us](#)

Flavoured Milk

Anchor's CalciYum low fat flavoured milk is an easy way to get dairy goodness into your week.

Just right for active growing kids - with loads of calcium to help build strong bones and natural dairy protein for growth and development as part of a healthy varied diet.



Featured

CalciYum Chocolate
Flavoured Milk →



Featured

CalciYum Strawberry
Flavoured Milk →

Flavoured Milk...?



Advertising Standards Authority



The screenshot shows the ASA website with a grey header containing the ASA logo and the text "Advertising Standards Authority". Below the header is a dark navigation bar with links for "Complaints", "Codes", "Decisions", and "AdHelp". The main content area has a blue banner with the text "Every ad a responsible ad". Below this is a white box with the heading "ASA Celebrates 50 Years" and a paragraph: "The ASA is celebrating its 50th year of supporting responsible advertising in New Zealand. Learn more about what we've accomplished over that time." To the right of the text is a circular graphic with "50 YEARS" in the center, surrounded by blue arrows pointing outwards, with the ASA logo at the bottom.

ASA Advertising Standards Authority

Complaints Codes Decisions AdHelp

Every ad a responsible ad

ASA Celebrates 50 Years

The ASA is celebrating its 50th year of supporting responsible advertising in New Zealand. Learn more about what we've accomplished over that time.

50 YEARS
ASA

Fair Trading Act 1986

Version as at 10 July 2023



Fair Trading Act 1986

Public Act 1986 No 121

Date of assent 17 December 1986

Commencement see section 1(2)

Note

The Parliamentary Counsel Office has made editorial and format changes to this version using the powers under [subpart 2](#) of Part 3 of the Legislation Act 2019.

Note 4 at the end of this version provides a list of the amendments included in it.

This Act is administered by the Ministry of Business, Innovation, and Employment.

Milk

- Kids
- Late teens/early 20s
 - Male
- What next?
 - Further differentiation?
 - Market segments?

The Promotional Mix

- Advertising

Advertising

Television

📺 Videos :



iPhone 15 Storage | Don't Let Me Go | Apple

YouTube · Apple
19 Mar 2024



Introducing iPhone 15 | WOW | Apple

YouTube · Apple
13 Sept 2023



iPhone 15 Check In | New Driver | Apple

YouTube · Apple
21 Jan 2024



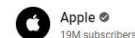
iPhone 15 Plus | Miss You | Apple

YouTube · Apple
25 Dec 2023

Online



iPhone 15 Storage | Don't Let Me Go | Apple



Subscribe

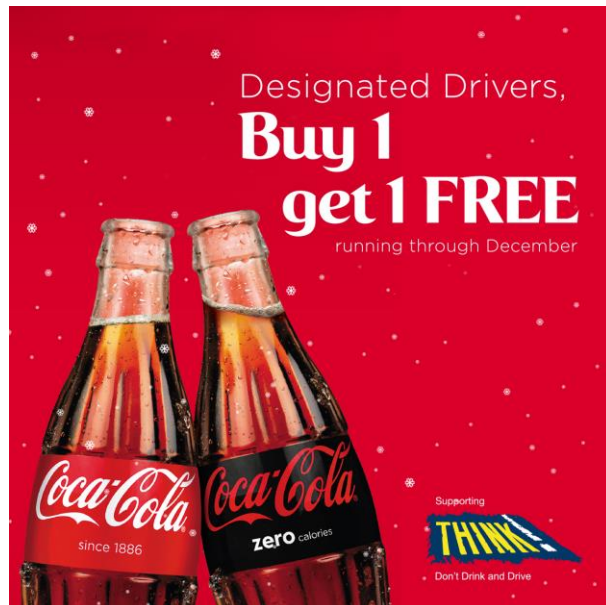
👍 28K 🗨️ 🔄 Share 📌 Save ⋮

- Advertising
 - Television
 - Online
 - Radio
 - Magazines
 - Radio
 - Billboards



The Promotional Mix

- Advertising
- Sales promotion



The Promotional Mix

- Advertising
- Sales promotion
- Public relations

Public Relations

- Deliberate effort by an organisation to create goodwill
- Methods can include:
 - Publicity – activities to promote a company/product via media, **not paid for by sponsor**
 - Sponsorship
 - Newsletters, magazines, annual reports, whitepapers
 - Tradeshow
 - Conferences
 - Lobbying

The Promotional Mix

- Advertising
- Sales promotion
- Public relations
- Personal selling

Personal Selling



The Promotional Mix

- Advertising
- Sales promotion
- Public relations/publicity
- Personal selling
- Product placement
- Digital marketing
- ...

Marketing

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